

# Kepler's Second Law by Newton in *Principia*

Rodrigo Souza Frausino  
rodrigofrausino6@gmail.com

## I. INTRODUCTION

Johannes Kepler formulated his laws of planetary motion based on astronomical observations, but their true foundation was not understood until Isaac Newton provided a mathematical proof. Using geometric arguments and his formulation of calculus, Newton demonstrated that these laws follow naturally from his laws of motion and universal gravitation.

In this discussion, we will examine Newton's proof of Kepler's Second Law as presented in his *Philosophiæ Naturalis Principia Mathematica* [1], showing how a planet's motion around the Sun follows precise geometric principles. Since that book was written in neo-latin, I found [2] to be a nice presentation about this proof; we follow it with added detail. We are interested in proving the following theorem

**Theorem 1.1** (Kepler's Second Law): A line segment joining a planet and the Sun sweeps out equal areas in equal time intervals.

In graphical terms, see the next diagram:

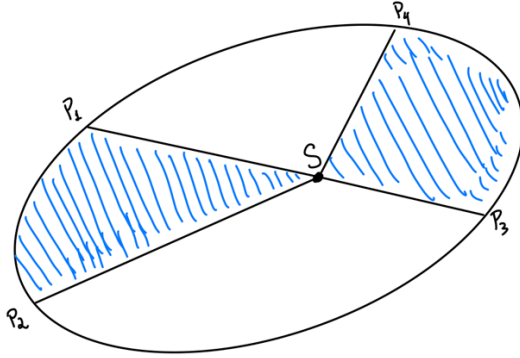


Fig. 1: Kepler's Second Law of planetary motion.

If the time it took from  $P_1$  to  $P_2$  is the same as  $P_3$  to  $P_4$ , then the highlighted areas are the same and vice-versa.

## II. NEWTON'S PROOF OF EQUAL AREAS

Newton's proof involves decomposing the elliptical motion into a sequence of small triangular areas, as illustrated in the next diagram:

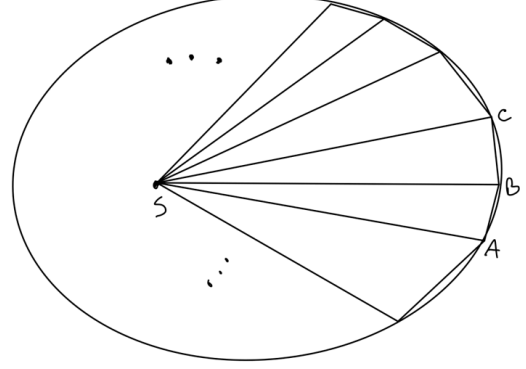


Fig. 2: Newton's idea to decompose in multiple triangles.

Let's assume that the motion is correctly depicted such that the time taken to travel from  $A$  to  $B$  is equal to the time taken from  $B$  to  $C$ , and so on. The theorem is proven once we establish that each of these triangles has the same area.

This is crucial for three reasons:

- 1) It confirms that equal time intervals correspond to equal areas, aligning with Kepler's second law, for these triangles.
- 2) As the time intervals become infinitesimally small, the sequence of triangles becomes a good approximation of the continuous elliptical motion.
- 3) Two regions swiped in equal time intervals have the same number of triangles contained within it (in the limit when the triangle becomes small enough), implying they have same areas since each triangle has the same area.

Thus, we need to prove that:

$$\text{Area}(\triangle SAB) = \text{Area}(\triangle SBC) \quad (1)$$

### A. Physical Intuition

If the planet starts at point  $A$  with an initial velocity  $v_0$  directed toward  $B$ , it will move in a straight line until reaching  $B$ . At this point, the gravitational force exerts an impulse that alters its velocity, changing its direction toward  $C$ . When the time steps are sufficiently small, this discrete change in direction closely approximates the effect of a continuous gravitational force acting throughout the elliptical motion.

### B. Proof

Zooming into two of the triangles in Fig. 2, we have:

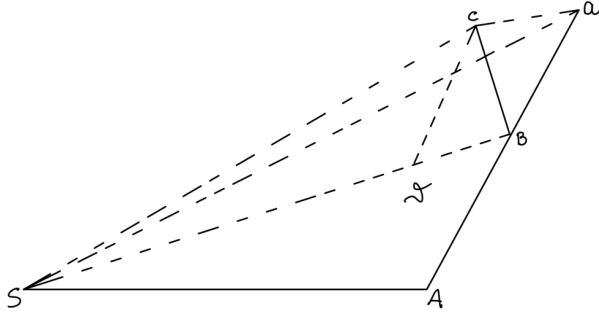


Fig. 3: Highlighted triangles for analysis.

Point  $a$  is the position where the planet would have been if there were no gravitational force acting on it—moving purely by inertia—in the same time interval that it would have reached  $C$ . Point  $v$  is the intersection of line  $SB$  and the line parallel to  $AB$  that passes through  $C$ .

It is evident that the segment  $AB$  is equal in length to  $Ba$ , since the planet's velocity would have remained constant in the absence of external forces, and we are considering equal time intervals.

Now, consider the triangle  $\triangle SAa$ . The segment  $SB$  is the median of this triangle, meaning it divides the triangle into two regions of equal area. This proves that:

$$\text{Area}(\triangle SAB) = \text{Area}(\triangle SBA). \quad (2)$$

Our next step is to show that the quadrilateral  $vBaC$  is a parallelogram. By construction, the segments  $vC$  and  $Ba$  are parallel, so we only need to establish that  $vB$  and  $Ca$  are also parallel.

At point  $B$ , the planet experiences an impulse that adds a velocity component  $v_1$  directed toward  $S$ . This results in a resultant velocity  $v_r$  directed toward point  $C$ , as illustrated in the next figure:

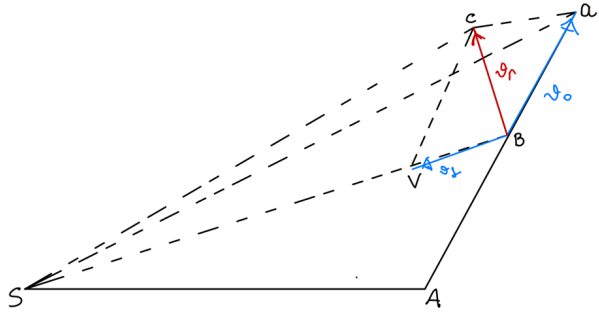


Fig. 4: Velocities satisfying the parallelogram law.

By the parallelogram law for vector addition, this implies that  $Ca$  and  $vB$  must be parallel. Consequently, we can conclude that:

$$\text{Area}(\triangle SBC) = \text{Area}(\triangle SBA). \quad (3)$$

This follows because both triangles share the base  $SB$ . Moreover, since  $vB$  and  $Ca$  are parallel, the perpendicular distance from  $C$  to  $SB$  is the same as the distance from  $a$  to  $SB$ . Therefore, the areas of the two triangles are equal.

This final result implies equation (1) and, consequently, proves Kepler's Second Law.

### III. CONCLUSION

This is an elegant and insightful proof of Kepler's Second Law, remarkable in that it does not require the modern tools of calculus. Newton's geometric approach captures the essence of gravitational motion through purely classical reasoning.

I wanted to share this proof with the reader and encourage anyone interested in physics—especially those who appreciate mathematical rigor—to explore Spivak's book [2]. Unfortunately, due to his passing, the series, which as far as I know was intended to cover electromagnetism, special relativity, and much more, will remain incomplete.

Nevertheless, Spivak's contributions to continue to inspire generations of students and researchers. And in my case, it allowed me to witness the proof of a legendary physicist and mathematician—the great Isaac Newton—through a fresh lens.

### REFERENCES

- [1] I. Newton, *Philosophiae naturalis principia mathematica*, vol. 1. 1687.
- [2] M. Spivak, *Physics for Mathematicians: Mechanics I*. Publish or Perish, 2010.